CS 289A Fall 2026

Grad Project Progress

Matthew Bernard, Shiqi Yang *November 6, 2026*

Progress summary

The experimental design is as follows: image-only ScienceQA (`lmms-lab/ScienceQA-IMG`) for main experiments and text-only as secondary analysis; the softmax-over-options variant as an additional ablation alongside $k = 8$ sampling; oracle and $\tau = 0$ baselines for the consistency-filter comparison; VQAv2 as a stretch goal only. All six action items were completed.

Action item	Owner	Status
Confirm SCF GPU allocation	Matt	Done — TITAN Xp (12 GB), CUDA 12.8, `PreemptMode=REQUEUE`
Set up Python env; confirm SmolVLM loads	Matt	Done — conda `smolvlm` (Python 3.11), torch 2.14.1+cu126; 507.48M params at bf16
HF Hub repo for checkpoints	Matt	Done — `mattbix/cs289-smolvlm-checkpoints`
Zero-shot greedy baseline, 500-question subset	Matt	Done — 0.6480 (324/500)
ScienceQA counts, image+text vs. text-only, per split	Matt, Shiqi	Done — Matt ran, Shiqi independently reproduced
Softmax-over-options API notes	Matt, Shiqi	Done — `02_softmax_eval.py`; Shiqi's check found the read-position issue

Two infrastructure notes. The proposal assumed an A100 or L4 (24 GB); the available GPU was a single TITAN Xp (12 GB), and SmolVLM-500M's ~1.1 GB bf16 footprint meant LoRA fit comfortably, so the proposal's assumption did not need to be met. The `PreemptMode=REQUEUE` partition required per-example checkpointing, which both baseline scripts did (JSONL append after every question); and the HF Hub token, stored under `$HOME`, survived login-node changes, with `HF_HOME` exported from `~/.bashrc` and `~/.profile` to cover both `srun` and `sbatch`.

Preliminary results

Dataset counts

Dataset	Split	Total	Image+text	Text-only
`lmms-lab/ScienceQA-IMG`	train	6,218	6,218	0
	validation	2,097	2,097	0
	test	2,017	2,017	0
`derek-thomas/ScienceQA`	test	4,241	2,017	2,224

The `derek-thomas/ScienceQA` image+text column matched `lmms-lab/ScienceQA-IMG` row-for-row on the test split. The main experiments therefore used `lmms-lab/ScienceQA-IMG`; the text-only rows were held out for a secondary analysis.

Baseline (500-question image-only test subset, seed 0)

Variant	Accuracy	Forward passes per question
Zero-shot greedy	0.6480 (324/500)	1 autoregressive decode, ≤ 8 tokens

The zero-shot greedy baseline was evaluated on a 500-question random subset of the image-only test split, seed 0 so the subset is reproducible; for the fine-tuning experiments, the validation split would be used for tuning and the test split reserved for final evaluation.

Softmax variant — read-position issue, and no significant difference from greedy

The initial softmax script read the answer distribution at the last position of the input prompt, immediately after the literal `"Answer:"` string. Shiqi's independent check found that at that position the model's top token was `" Answer"` on 48 of 50 questions, and the option letters held only a small fraction of the probability. That read position measured a different decision point, not a better-calibrated one, so the 0.6700 figure from the initial 500-question run was withdrawn. Shiqi's corrected version, which prefills `" Answer:"` in the assistant turn and reads the logits after that, agreed with greedy on 48 of 50 questions in his 50-question check. Separately, the initial 59-vs-48 greedy/softmax disagreement count was not statistically significant (McNemar $p \approx 0.3$), so neither finding supports the "softmax captures information greedy misses" reading. The $k = 8$ sampling variant has not yet been run; the "complementary signals" question depends on that comparison and is deferred to the post-Nov-6 experiments.

Updated timeline

Week	Milestone	Status
Nov 6	**Progress Report due.**	Submitted
Nov 7–13	$k = 8$ sampling baseline; corrected-softmax full-500 run if needed.	Planned
Nov 14–20	Consistency-filter fine-tuning: threshold τ, oracle + $\tau = 0$ baselines.	Planned
Nov 21–24	**First Draft of Paper due Nov 24.**	Planned
Nov 25–Dec 3	Ablations: k, τ, T, LoRA rank, epochs. Error-propagation second round.	Planned
Dec 4	**Peer Review due Dec 4.**	Planned
Dec 5–11	Final revisions; record video. **Final Report + Video due Dec 11.**	Planned

Remaining work, outline, and code

Beyond the timeline above: the corrected softmax full-500 run (optional), the $k = 8$ sampled-mode baseline, the consistency-filter fine-tuning with oracle and $\tau = 0$ comparisons, SmolVLM pseudo-label generation on the training split, LoRA fine-tuning with validation-tuning and held-out test evaluation, and the post-fine-tuning re-run of both baselines. The single SCF TITAN Xp was sufficient for SmolVLM-500M inference and LoRA fine-tuning; the course's $300 in Databricks credits were identified as a backup for longer runs or parallelism if the SCF partition became contended.

Paper outline: outline.pdf

Code: github.com/mattbix/cs289gradproject. Shiqi's independent checks are on the `shiqi/independent-checks` branch, with notes in `notes/softmax_api_notes.md`. Checkpoints: huggingface.co/mattbix/cs289-smolvlm-checkpoints.